

Architecture undergraduate · Iran University of Science and Technology

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AREAS OF WORK

Computational design & spatial systems · Digital fabrication & robotic assembly

Architecture & building data · Multimodal human–environment research

01 / EDUCATION

Iran University of Science and Technology (IUST)

Expected 2027

Bachelor's degree in Architecture · In progress

Thesis: Lunar research station · Supervisor: Dr. Morteza Rahbar

Secondary education: Allameh Helli 1, Tehran · NODET / SAMPAD

02 / RESEARCH EXPERIENCE

Research Assistant | Multimodal human–environment research

2026–present

Dr. Boshra Abbasi · EEG–VR laboratory work at IUST

- Designed a **reusable research platform** integrating VR/360° experiences with EEG, event, spatial, behavioural and questionnaire records.
- Implemented Lab Streaming Layer (LSL) synchronisation and session logging; conducted **64-channel EEG–VR pilots** to assess signal quality, movement artefacts and headset–cap integration.
- Prepared experimental protocols, evidence reviews, pilot analyses and research documentation.

03 / SELECTED SOFTWARE, DESIGN & FABRICATION

Residence Hell | Computational residential design software

Ongoing

System author & architectural designer · Independent academic project

- Designed configurable residential design software using voxel/graph representations, multi-agent growth, environmental evaluation, pathfinding and **multi-objective/Pareto search**.
- Integrated massing, floor-plan synthesis and IFC export with **editable states, branching** and comparison of design alternatives.
- Authored system architecture, spatial rules and evaluation criteria; directed AI-assisted implementation, testing and revision.

Wood–Thread | Full-scale parametric pavilion

May 2025

Designer & parametric modeller · Harakat Festival, IUST

- Designed and parametrically modelled a wood-and-thread pavilion (approx. **5 × 2 × 2.1 m**) for the Architecture Scientific Association's festival booth.
- Produced **CNC fabrication files** and collaborated in fabrication and on-site assembly.

Nefar | Vernacular construction & robotic assembly

2026

System author & designer · Robotic assembly software and simulation

- Translated field surveys and structural models of Alam Kola timber construction into discrete members, joints and **assembly dependencies**.
- Built Python-based **robotic assembly software** integrating inverse kinematics, reachability, grasp targets and collision-aware pick–carry–place sequences.
- Modelled robot/gantry cells and tool frames; used simulation to refine assembly order, component handling and temporary-support logic.

04 / PROFESSIONAL EXPERIENCE

- Municipal museum | Architectural design collaboration

Design team with Dr. Abbaszadeh and Dr. Tafti · Municipal project, Tehran

Developed architectural concepts, site/context analysis, digital models and 3D-printed study models for a municipal museum planned for construction.

2026
- Tehran Experience Studio | Construction-site intern

Documented construction progress and maintained site checklists, relating architectural drawings to on-site execution.

Aug-Sep 2024
- Hermapay Studio | Architectural design intern

Produced digital models and project documentation; surveyed existing conditions in Tehran's historic Naseri urban fabric.

Jul-Sep 2023
- Independent 3D-printing services

Provided FDM printing for external projects, including model preparation, slicing and print production.

Project-based

05 / ADDITIONAL DESIGN & TECHNICAL WORK

- Architectural design & analysis

Developed site/programme studies, plans, sections, spatial models and BIM documentation; explored light, ventilation and landscape integration across housing, museum, school and aquatic-centre work.
- Building-data tools

Prototyped IFC-to-Neo4j graphs and natural-language-to-Cypher queries; explored geometry voxelisation, spatial relationships and A* routing through building models.
- Learning & retrieval experiments

Ran building-code retrieval and local-model fine-tuning experiments; developed a self-play learning agent for a trick-taking card game.
- Physical systems & prototyping

Designed and 3D-printed adjustable ball-and-socket frames and bracing connections; developed parametric/laser-cut components, folded forms, joints and solid-modelled mechanisms.

06 / ACADEMIC ENGAGEMENT & TEACHING

- Architecture Scientific Association, IUST | Member

Contributed to event ideation, planning and execution, including logistics, on-site coordination and spatial setup across association activities.
- Conversation with Juhani Pallasmaa | Co-moderator

Co-hosted an IUST discussion on architecture, sensory experience, education and technology.

Jun 2026
- Concept-to-form architecture workshops | Instructor

Taught students in grades 7–8 across multiple workshop sessions, guiding concept development and translation into architectural form.

2026
- AI in architecture workshops

Practical AI applications in architectural ideation, visualisation and design workflows.

07 / METHODS & TOOLS

- Computation

Python · generative design · optimisation · building-data tools
- Robotics

ROS 2 · Isaac Lab · MuJoCo · kinematics · motion planning
- Design & BIM

Rhino · Grasshopper · Revit · Dynamo · AutoCAD · SolidWorks
- Fabrication

CNC · laser cutting · 3D printing · full-scale assembly
- Research systems

EEG/ERP · VR · LSL · multimodal recording & synchronisation
- Languages

Persian — native · English — working proficiency